

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2019

EE 2113

(Electrical Machines)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Mention four differences between electric and magnetic circuit. Explain leakage flux, fringing flux and magnetic field with proper diagram. 4+3×2=10
- 1(b). What do you mean by magnetizing curve and toroidal magnetic circuit? Draw the analogous electrical circuit for the circuit show in fig.1(b) 2+2+6=10

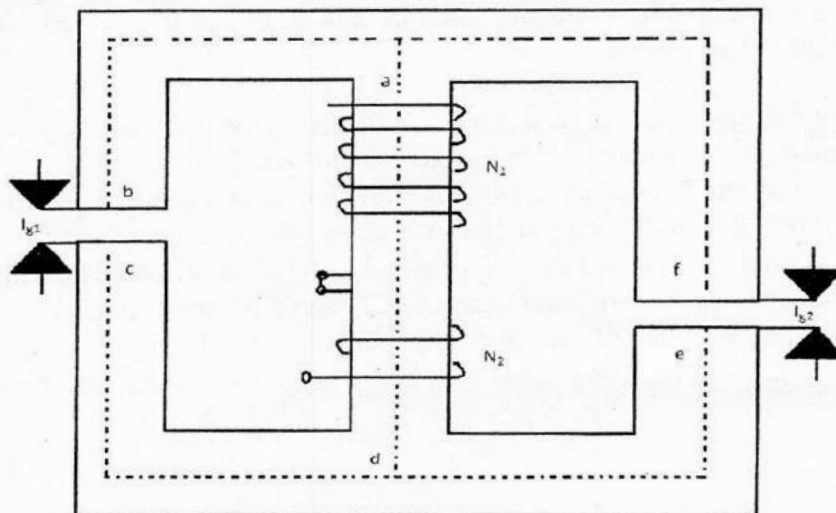


Fig.1(b)

- 1(c). Explain Ampere's circuital law. In the magnetic system of fig. 1(c) two sides are thicker than the other two sides. The depth of the core is 10 cm, the relative permeability of the core, $\mu_r = 2000$, the number of turns $N = 500$ and the current flowing through the coil is $i = 1A$, (i) Determine the flux in the core, (ii) Determine the flux densities in the part of the core. (iii) Find the current i in the coil to produce a flux ($\Phi = 0.012 \text{ wb}$) 2+4+4+5=15

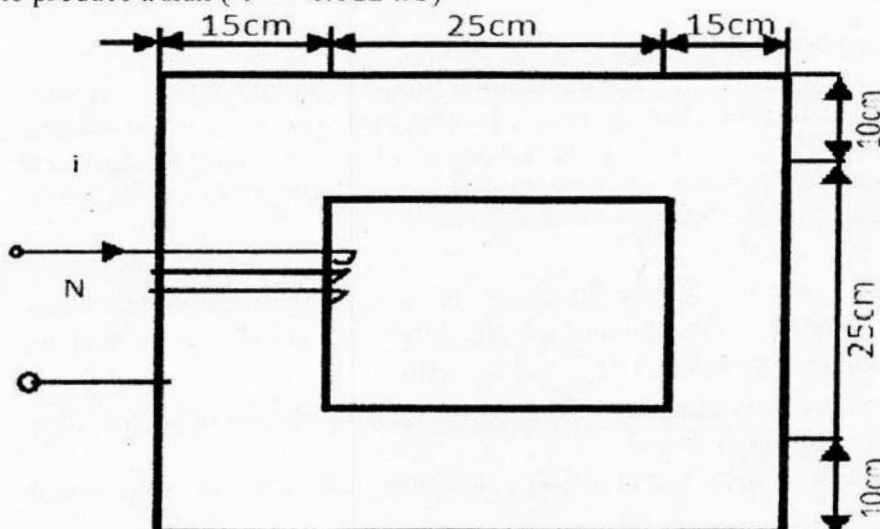


Fig. 1(c)

- 2(a). Write down the conditions for parallel operation of DC generator. Explain pole cores, pole coils, and pole pitch. 6+3=9
- 2(b). How the AC voltage in the armature winding is converted to DC in the brushes of generator? Explain with proper diagram. 08
- 2(c). Derive the conditions of maximum efficiency of a dc machine. and Show that $\eta_c = \eta_m * n_e$, where the symbols have their usual meanings. 10
- 2(d). A 4 pole dc generator runs at 750 rpm and generates an emf of 240 V. The armature is wave wounded and has 792 conductors. If the total flux from each pole is 0.0145 Wb, what is the leakage co-efficient? 08
- 3(a). Write down the specification of a typical DC motor . Also mention some applications of DC series and shunt motor. 08
- 3(b). "DC shunt motor is constant speed motor"- justify the statement. 08
- 3(c). How the speed of a dc motor can be controlled? Explain the principle of stater along with its necessity. 12
- 3(d). A dc series motor , connected to a 440V supply, runs at 600 rpm when taking a current of 50A. Calculate the value of resistance of which when inserted in series with the motor will reduce the speed to 400 rpm, the gross torque being then half its previous value. The resistance of the motor is 0.2 Ω . Assume flux is proportional to field current. 07
- 4(a). Define critical speed and crtical field resistance. Is voltage buildup process possible fan dc shunt generator? If possible, explain briefly. 10
- 4(b). Write down the significance of counter emf. Show that the torque produced by dc motor is proportional to the main field flux and armature current. 10
- 4(c). A shunt generator delivers 195A at terminal voltage of 250V. The armature resistance and shunt field resistance are 0.02 Ω and 50 Ω respectively. The iron and friction losses equal 950W. Find (i) EMF generated. (ii) Cu losses (iii) Output of the prime mover (iv) Commercial, mechanical and electrical efficiency. 15

SECTION – B

- 5(a). Describe working principle of transformer on load. Also deduce the *E.M.F.* equation of a transformer. 6+5=11
- 5(b). Explain the procedures and outcomes of open-circuit and short-circuit test of a transformer. 5x2=10
- 5(c). Draw the exact equivalent circuit of a transformer. 06
- 5(d). A 30KVA, 2400/120V, 50 Hz transformer has a high voltage winding resistance of a 0.1 Ω and a leakage reactance of 0.22 Ω . The low voltage winding resistance is 0.035 Ω and the leakage reactance is 0.012 Ω . Find the equivalent winding reactance, reactance and impedance referred to the (i) high-voltage side and (ii) the low- voltage side 08
- 6(a). What do you mean by 'vector grouping' of a 3 phase transformer? Show primary and secondary connection of a 3 phase transformer with vector diagram for the following vector group: (i) Y_{y6} , (ii) Y_{d11} , (iii) D_{d6} . 2+3x3=11
- 6(b). Why induction motor is named so? Write down the applications of single phase and three phase induction motor. 08
- 6(c). What is slip of an induction motor? Why does the rotor rotate in an induction motor? 1+5=6
- 6(d). A 25KVA transformer has 500 turns on the primary and 50 turns on the secondary winding. The primary is connected to 3000 V, 50 Hz supply. Find the full-load primary and secondary currents, the secondary *e.m.f.*, and the maximum flux in the core. Neglect leakage drops and no-load primary current. 3+2+2+3=10

- 7(a). "Maximum torque can be achieved at any speed in a slip-ring induction motor"- Justify this statement by deriving torque equation and maximum torque condition under running condition of an induction motor. 8+5+2=15
- 7(b). Show that, a 3 phase supply to an induction motor is sufficient to produce necessary rotating magnetic field. 07
- 7(c). Show that power stages in an induction motor. Explain complete torque/speed curve of a three phase machine. 2+3=5
- 7(d). A 3-phase, slip-ring induction motor with star-connected rotor has an induced e.m.f of 120V between slip-rings at standstill with normal voltage applied to the stator. The rotor winding has a resistance per phase of 0.3 ohm and standstill leakage reactance per phase of 1.5 ohm. Calculate, (i) rotor current/phase when running short-circuited with 4% slip and (ii) the slip and rotor current per phase when the rotor is developing maximum torque. 3+5=8
- 8(a). Describe pitch factor and synchronous reactance. Deduce the equation for actual induced voltage/phase in an alternator. 2+5=7
- 8(b). What is voltage regulation of an alternator? Mention some methods to determine voltage regulation. Explain one of the methods in detail. 1+1+6=8
- 8(c). What is armature reaction? Explain the effect of armature reaction in an alternator for different loads. 2+8=10
- 8(d). In a 50KVA, star-connected, 440V, 3 phase, 50 Hz alternator, the effective armature resistance is 0.25Ω / phase. The synchronous reactance is 3.2 ohm per phase and leakage reactance is 0.5 ohm per phase. Determine at rated load and unity power factor: (a) Internal e.m.f. E_a (b) no-load e.m.f. E_0 , (c) Percentage regulation on full-load, (d) value of synchronous reactance which replaces armature reaction. 3+3+2+2=10

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2019

Math 2113

(Linear Algebra and Vector Analysis)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$, where t is time. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\hat{i} - 3\hat{j} + 2\hat{k}$. 12
- 1(b). Find the unit tangent vector to the curve $x = t^2 + 1$, $y = 4t - 3$, $z = 2t^2 - 6t$, at the point $t = 2$. 09
- 1(c). A particle moves along the space curve $x = 3 \cos t$, $y = 3 \sin t$, $z = 4t$. Find the curvature κ and torsion τ . 14
- 2(a). Define level curve, graph, and surface. Consider a function $f(x, y) = 100 - x^2 - y^2$. Find the graph and plot the level curves $f(x, y) = 0$, $f(x, y) = 51$, $f(x, y) = 75$ in the domain of f in the plane. 12
- 2(b). Find the equation of tangent plane and normal line to the surface $z = x^2 + y^2$ at the point $(1, -1, 2)$. 14
- 2(c). Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at $(2, -1, 2)$ in the direction $2\hat{i} - 3\hat{j} + 6\hat{k}$. 09
- 3(a). Define solenoidal vector and conservative vector field. Prove that the vector $\underline{A} = (2x^2 + 8xy^2z)\underline{i} + (3x^3y - 3xy)\underline{j} - (4y^2z^2 + 2x^3z)\underline{k}$ is not solenoidal but $\underline{B} = xyz^2\underline{A}$ is solenoidal. 11
- 3(b). If $\underline{F} = (2x + y)\underline{i} + (3y - x)\underline{j}$, evaluate $\int_c \underline{F} \cdot d\underline{r}$ where c is the curve in the xy -plane consisting of the straight lines from $(0,0)$ to $(2,0)$ then to $(3,2)$. 12
- 3(c). Verify the divergence theorem for $\vec{A} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ bounded by the region $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, and $z = 1$. 12
- 4(a). If $\underline{v} = \underline{w} d\underline{r}$, prove that $\underline{w} = \frac{1}{2} \text{curl } \underline{v}$ where $\underline{w}$ is a constant vector. 10
- 4(b). Evaluate $\iint_S \underline{A} \cdot \underline{n} ds$ where $\underline{A} = y\underline{i} + 2x\underline{j} - z\underline{k}$ and S is the surface of the plane $2x + y = 6$ in the first octant cut off by the plane $z = 4$. 13
- 4(c). Verify Stokes theorem for $\vec{A} = (2x - y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$, where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary. 12

SECTION – B

- 5(a). Define with example (i) elementary matrix, (ii) principal submatrix, and (iii) skew-symmetric matrix. 09
- 5(b). Define Inverse matrix. Find the inverse of $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ by using adjoint and show that $A^{-1} = A^2$. 12
- 5(c). What is consistence and inconsistency? For what values of λ the system of linear equations $x + y + z = 1$, $x + 2y + 4z = \lambda$, $x + 4y + 10z = \lambda^2$ has a solution and solve them completely in each case. 14
- 6(a). The set $\{1, t, e^t, te^t\}$ is a basis of a vector space v of functions $f: R \rightarrow R$. Let D be the differential operator on v . That is, $D(f) = \frac{df}{dt}$. Find the matrix D in the given basis. 11
- 6(b). Reduce the following matrix A to its echelon form then to its canonical form. Also, find its rank. 12
- $$A = \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 2 & -5 & 3 & 1 \\ 4 & 1 & 1 & 5 \end{bmatrix}$$
- 6(c). Consider the mapping $A: R^4 \rightarrow R^3$ where, $A = \begin{pmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 5 & -2 \\ 3 & 8 & 13 & -3 \end{pmatrix}$. 12
- Find a basis and dimension of (i) image of A , and (ii) Kernel of A .
- 7(a). Define vector space. Let W be the subspace of R^5 spanned by the vectors $(1, -2, 0, 0, 3)$, $(2, -5, -3, -2, 6)$, $(0, 5, 15, 10, 0)$, and $(2, 6, 18, 8, 6)$. Find the basis and dimension of W . 12
- 7(b). State Cauchy – Schwartz inequality. Verify that the following is an inner product in R^2 , $\langle u, v \rangle = x_1y_1 - x_1y_2 - x_2y_1 + 3x_2y_2$, where $u = (x_1, x_2)$, and $v = (y_1, y_2)$. 13
- 7(c). Write the properties of linear transformation. User matrix multiplication to find the image of vector $(2, -1, 2)$ if it is rotated (i) 30° clockwise about the positive x-axis, and (ii) 30° counterclockwise about the positive y-axis. 10
- 8(a). Find an orthogonal and an orthogonal basis for the subspace v of R^4 spanned by the vectors $v_1 = (1, 1, 1, 1)$, $v_2 = (1, -1, 2, 2)$, and $v_3 = (1, 2, -3, -4)$. 10
- 8(b). Find the reflection of the vector $x = (1, 5)$ about the line through the origin makes an angle of $\frac{\pi}{6}$ with the x-axis. 10
- 8(c). Find the eigen values and corresponding eigen vector of the matrix, 15

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2019

ME 2113

(Statics and solid mechanics)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Cable AC exerts a force P on beam AB directed along line AC. Knowing that P must have a 1750 N vertical component, determine (i) the magnitude of the force P, (ii) its horizontal component, (iii) what would be the value of P if angle CAB is changed to 20° under the same condition above? [See Fig. Q. 1(a)] 17

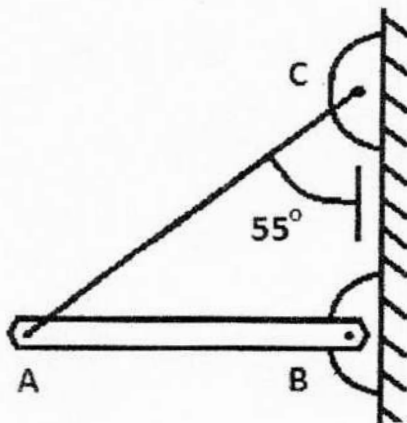


Fig. Q. 1(a)

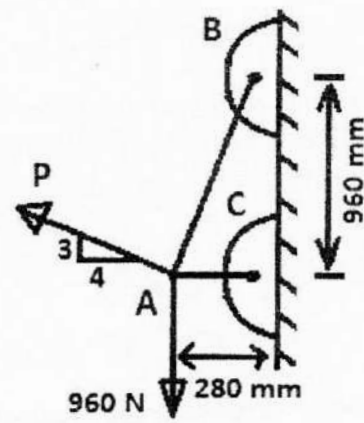


Fig. Q. 1(b)

- 1(b). Two cables are tied together at A and loaded as shown. Knowing that $P = 640$ N, determine the tension in each cable. Also, determine the value of P for which the tension in cable AB becomes zero. [See Fig. Q. 1(b)] 18

- 2(a). The lever AB is hinged at C and attached to a control cable A. If the lever is subjected at B to a 500 N horizontal force, determine (i) the tension in the cable, (ii) the reaction at C. [See Fig. Q. 2(a)] 18

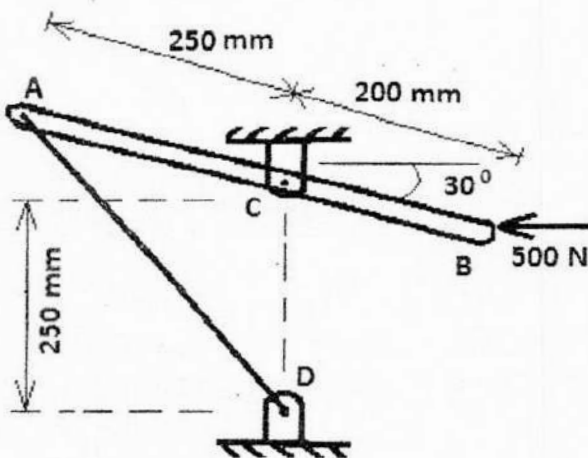


Fig Q. 2(a)

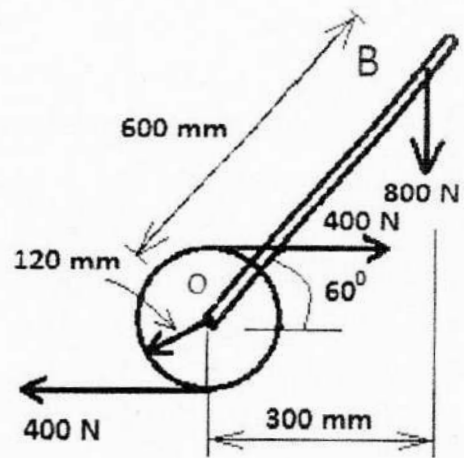


Fig. Q. 2(b)

- 2(b). Replace the couple and force by equivalent single force applied to the lever. Determine the distance from the shaft to the point of application of this equivalent force. [See Fig. Q. 2(b)] 17

- 3(a). Determine the centroid of the area shown in Fig 3(a) by direct integration method. [See Fig. Q. 3(a)] 17

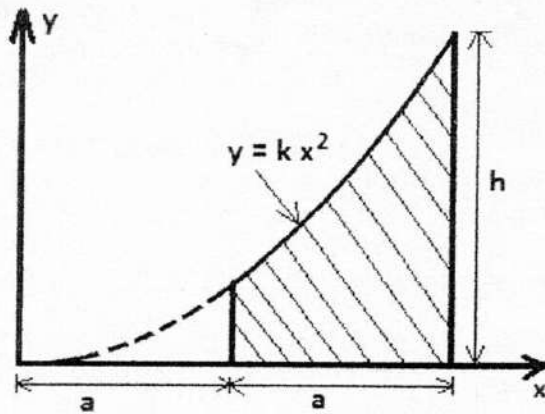


Fig. Q. 3(a)

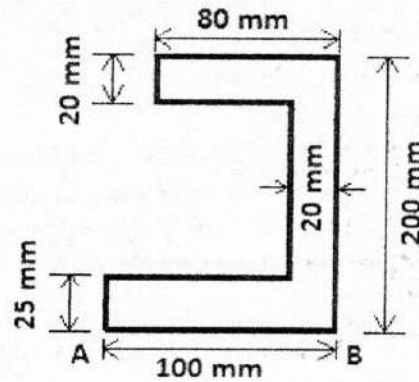


Fig. Q. 3(b)

- 3(b). Determine the moments of inertia $\bar{I}_x$ and $\bar{I}_y$ of the area shown with respect to centroidal axes respectively parallel and perpendicular to the side AB. [See Fig. Q. 3(b)] 18
- 4(a). For slipping belt, show that $T_2 = T_1 e^{\mu_k \beta}$ where the symbols have usual meaning. 17
- 4(b). The coefficients of friction between the block and rail are $\mu_s = 0.30$ and $\mu_k = 0.25$. Find the magnitude and direction of the smallest force P required (i) to start the block up the rail, (ii) to keep it from moving down. [See Fig. Q. 4(b)] 18

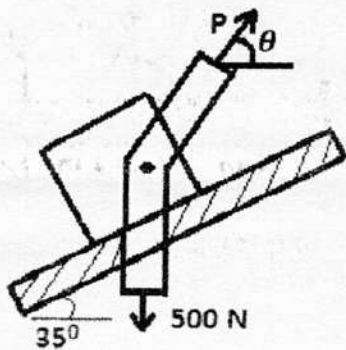


Fig. Q. 4(b)

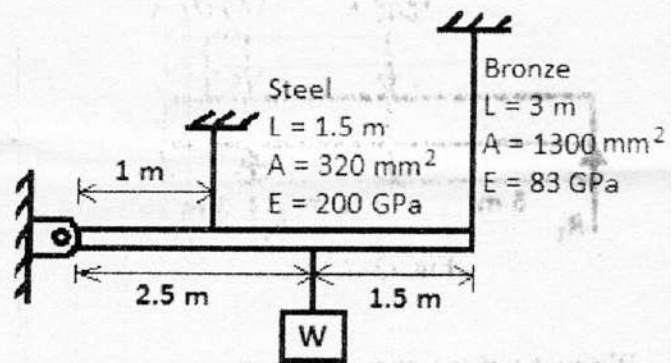


Fig. Q. 5(c)

SECTION - B

- 5(a). Define tensile stress, compressive stress, and bending stress. 03
- 5(b). Mathematically show that, if pressure in a cylinder is raised to the bursting point, failure will occur along a longitudinal section of the cylinder. 15
- 5(c). A rigid bar of negligible weight is supported as shown. If $W = 80$ kN, compute the temperature change that will cause the stress in the steel rod to be 55 MPa. Assume the co-efficient of linear expansion are $11.7 \mu\text{m}/(\text{m}^\circ\text{C})$ for steel and $18.9 \mu\text{m}/(\text{m}^\circ\text{C})$ for bronze. [See Fig. Q. 5(c)] 17

- 6(a). A compound shaft consisting of a steel segment and an aluminum segment is acted upon by two torques as shown. Determine the maximum permissible value of T subjected to the following conditions: $\tau_{st} \leq 83 \text{ MPa}$, $\tau_{al} \leq 55 \text{ MPa}$, and the angle of rotation of the free end is limited to 6° . For steel $G = 83 \text{ GPa}$ and for aluminum $G = 28 \text{ GPa}$. [See Fig. Q. 6(a)] 17

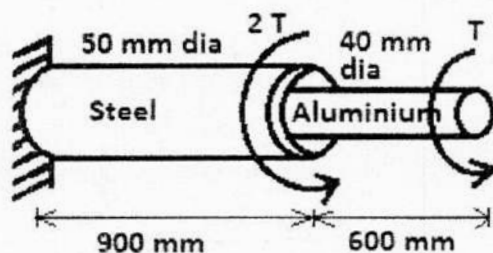


Fig. Q. 6(a)

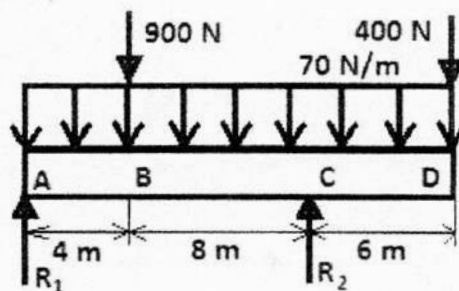


Fig. Q. 6(b)

- 6(b). Write the shear and moment equations for the beam loaded as shown. Also draw shear force and bending moment diagrams, specifying values at all change of loading points and at all points of zero shear. [See Fig. Q. 6(b)] 18
- 7(a). Derive the flexure formula stating the assumptions. 18
- 7(b). Compute the value of EI_y at the right end of the overhanging beam shown in figure using double integration method. [See Fig. Q. 7(b)] 17

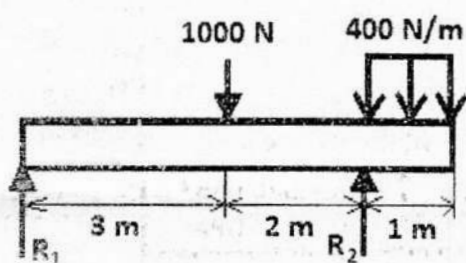


Fig. Q. 7(b)

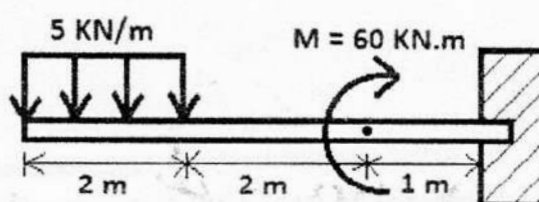


Fig. Q. 8(a)

- 8(a). Without writing shear or moment equations, sketch shear and moment diagrams for the beam as shown. Also, compute values at all change of loading points and the values of maximum shear and maximum moment. [See Fig. Q. 8(a)] 17
- 8(b). What is critical load of a column? Select the lightest W shape that will act as a column 8 m long with hinged ends and support an axial load of 270 kN with a factor of safety of 2.5. Assume that the proportional limit is 200 MPa and $E = 200 \text{ GPa}$. 18

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2019

ME 2115

(Fluid Mechanics - I)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Explain the terms: (i) Ideal fluid, (ii) Newtonian fluid, and (iii) Non-Newtonian fluid. 12
- 1(b). Explain the rheological diagram. 08
- 1(c). Describe Newton's law of viscosity. Show that, $\tau_{xy} = \mu \left(\frac{\partial u}{\partial y} \right)$, where symbols have their usual meaning. 15
- 2(a). Define atmospheric, absolute, gauge, and vacuum pressures. Also show the relationship among them. 10
- 2(b). Differentiate between: simple and differential manometer. Draw a schematic diagram of a single column manometer, inverted U-tube manometer, and U-tube differential manometer. 13
- 2(c). A shaft of 70 mm diameter rotates inside a fixed cylinder of 72 mm diameter. The length of cylinder 0.6 m. Find the rpm of the shaft if the space between the shaft and cylinder is filled with lubricating oil of viscosity 2.5 poise and a torque of 1.2 Nm is applied. Find the power required to rotate the shaft. [See Fig. Q. 2(c)] 12

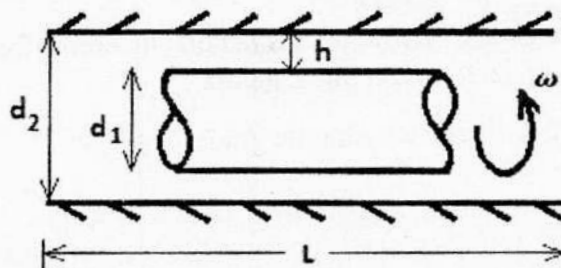


Fig. Q. 2(c)

- 3(a). Prove that the rate of increase of pressure in a vertical direction is equal to weight density of the fluid at that point. 10
- 3(b). Define total pressure and center of pressure. Derive the expression for total pressure on a vertical plane surface. 13
- 3(c). Show that the center of pressure is always below the centroid of the vertical submerged plane. 12
- 4(a). Differentiate between Lagrangian and Eulerian method of fluid motion. 06
- 4(b). Define the following terms: 09
(i) Streamline, (ii) Pathline, (iii) Streak line.
- 4(c). Derive Bernoulli equation along a streamline. What are the necessary assumptions needed to apply this equation to a fluid dynamic problem? 10

- 4(d). The water in a large lake is to be used to generate electricity by the installation of a hydraulic turbine generator. The elevation difference between the free surfaces at upstream and downstream of the dam is 50 m. Water is to be supplied at a rate of 5000 Kg/s. If the electric power generated is measured to be 1862 KW and the generator efficiency is 95 percent, determine (a) the overall efficiency of the turbine – generator, (b) the mechanical efficiency of the turbine, (c) the shaft power supplied by the turbine to generator. [See Fig. Q. 4(d)]

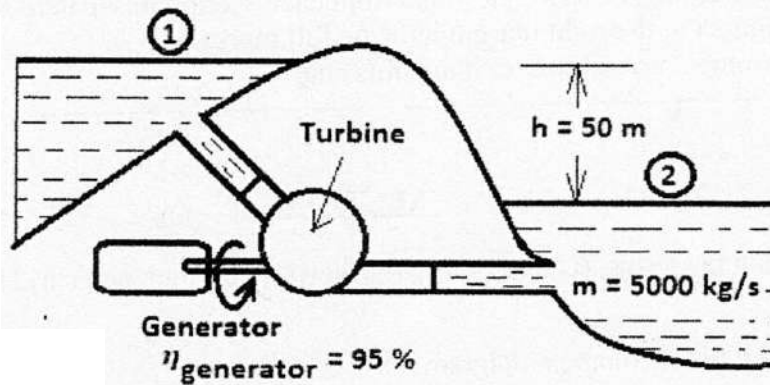


Fig. Q. 4(d)

SECTION – B

- 5(a). Derive Euler equation for fluid flow using material derivative from Newton's law, $\sum \vec{F} = m \vec{a}$. 15
- 5(b). Derive the expression of continuity equation in three dimension. 15
- 5(c). Write short notes on: (i) internal flow, (ii) external flow. 05
- 6(a). What are the basic approaches to analyze the arbitrary fluid flow problems? Discuss them briefly. 07
- 6(b). Deduce one-dimensional Reynolds transport theorem for a fixed volume, with the physical meaning of each term in the equation. 20
- 6(c). What is energy grade line and hydraulic grade line? 08
- 7(a). What are meant by drag force and lift force? What are the factors on which drag and lift co-efficient depend? 10
- 7(b). Show that for high speed flows, pressure drag co-efficient is independent of Reynolds number, whereas for low speed flows it depends inversely with Reynolds number. 12
- 7(c). Discuss the salient features of steady flow past a circular cylinder with neat sketch for flows of very small, moderate, and very high Reynolds number. 13
- 8(a). What is hydrodynamic boundary layer? What are the factors that affect boundary layer thickness? 08
- 8(b). Derive the general integral momentum equation of the boundary layer, when pressure is constant through the flow. 17
- 8(c). A laminar boundary layer at zero pressure gradient over a flat plate is described by the velocity profile: 10

$$\frac{v}{V} = \sin \left(\frac{\pi y}{2 \delta} \right).$$

Determine the displacement thickness, momentum thickness, and energy thickness.

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2019

CSE 2113

(Computer Programming)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is GNU Octave? What is GNU stand for? Write the application and limitation of GNU Octave. Compare Octave and Matlab. 1+1+2+2
+3= 09
- 1(b). Write a program to take two matrixes A and B as input from user. A is a (1×3) matrix with complex numbers as its elements. B is a (3×1) matrix with complex numbers as its elements. Multiply A and B show the result and show real and imaginary parts of the result separately. 05
- 1(c). Write short notes on: i) [s i] = sort (A, dir, opt). ii) [i j] = find (A< 0.5). iii) C = floor (10.9) – fix (5.6). Mention what these functions do, what their input arguments and outputs are. 3*2=06
- 1(d). The electric field intensity, $E(Z)$, due to a ring of radius R at any point Z along the axis of the ring is given by: $E(Z) = \frac{\lambda}{2\epsilon_0} \frac{RZ}{(z^2 + R^2)^{\frac{3}{2}}}$, Here, λ is the charge density, $\epsilon_0 = 8.85 \times 10^{-12}$ is the electric constant, and R is the radius of the ring. Consider the case where $\lambda = 1.7 \times 10^{-7} \text{ cm}^{-1}$ and $R = 6 \text{ cm}$. Do the following:
i) Write an anonymous function Ez to determine $E(Z)$. Use it to determine $E(Z)$ at $Z = 0, 2, 4, 6, 8$ and 10 cm .
ii) Use the anonymous function Ez to determine the distance Z where E is maximum. Do it by creating a vector Z with elements ranging from 2 cm to 6 cm and spacing of 0.05 cm . Calculate E for each value of Z , and then find the maximum E and associated Z with Octave's built-in function 'max'. 7+8=15
- 2(a). Show two methods to create a structure named projectile, which have three fields: type, mass, and velocity. Create two objects of this structure with different values: projectile_1 and projectile_2. Show output of each command. Can we implement projectile_1 and projectile_2 using cell array instead of structure? If yes, show how it can be done. 4+2+2+1
+3=12
- 2(b). Discuss different ways to input value of a variable inside a script file. 08
- 2(c). Write Octave code for followings: 15
(i) Solving linear system
$$\begin{aligned} 4x - 2y + 6z &= 8 \\ 2x + 8y + 2z &= 4 \\ 6x + 10y + 3z &= 0 \end{aligned}$$

(ii) Create a random matrix with three rows and four columns. Pre and post multiply $v = [1, 2, 3, 4]$ with the previous matrix.
(iii) Use built-in function to calculate angle between $\vec{r}_1 = 6\hat{i} - 3\hat{j} + 2\hat{k}$ and $\vec{r}_2 = 2\hat{i} + 9\hat{j} + 10\hat{k}$.
- 3(a). Discuss different techniques to plot multiple graphs in the same plot. How can we put multiple plots on the same page? Explain. 9+3=12
- 3(b). Discuss on 'Histogram' and 'Polar plot'. Mention all possible input arguments and output of these functions. 3×2=6

- 3(c). Why do we need loops in programming? Show different loop mechanism of Octave with proper example. 2+6=8
- 3(d). The graph of the function $f(x) = ax^3 + bx^2 + cx + d$ passes through the points $(-2.6, -68)$, $(0.5, 5.7)$, $(1.5, 4.9)$, and $(3.5, 88)$. Write an Octave code for finding the constants a, b, c , and d . 09
- 4(a). Why do we need 'function' in programming? Mention differences between function and script file. What is the prototype of a function? 1+3+1=5
- 4(b). Write short notes on : (i) break and continue, (ii) anonymous function (iii) Passing a user-defined function into a function function.(iv) switch. 2×4=8
- 4(c). Write a function 'y= isPrime(x)' to check whether x is prime (y = 1 if x is prime) or not (y = 0 if x is not prime). Using this function write a program to find first 50 prime numbers. Store these prime numbers in variable - primeNumbers, and display these numbers. 6+6=12
- 4(d). The factorial $n!$ of a positive integer is defined by $n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 1$, where $0! = 1$. Write a user-defined function that calculates the factorial $n!$ of a number. For function name and arguments use $y = \text{fact}(x)$, where the input argument x is the number whose factorial is to be calculated and the output arguments y is the value $x!$. The function displays an error-message if a negative or non-integer number is entered when the function is called. Do not use Octave's built-in function 'factorial' 10

SECTION – B

- 5(a). Cam is a mechanical device that transfers rotary motion into linear motion. The motion of a cam is given by the following equations- 10
- $y = 6[2\theta - 0.5 \sin \theta]/\pi$; $0 \leq \theta < \pi/2$
 $y = 6$; $\frac{\pi}{2} \leq \theta < 2\pi/3$
 $y = 6 - 3[1 - 0.5 \cos 3(\theta - \frac{2\pi}{3})]$; $\frac{2\pi}{3} \leq \theta < 4\pi/3$
 $y = 6$; $\frac{4\pi}{3} \leq \theta < 3\pi/2$
- $y = 3 - 1.5 \left(\frac{\theta - 3(\frac{\pi}{2})}{\pi/4} \right)^2$; $\frac{3\pi}{2} \leq \theta < 7\pi/4$
 $y = 0.75 - 0.75 \left(\left(1 - \frac{\theta - 7(\frac{\pi}{4})}{\pi/4} \right)^2 \right)$; $\frac{7\pi}{4} \leq \theta < 2\pi$
- Write an Octave program that plots the displacement profile for one revolution. First create vector θ with 500 elements, then by using a loop and conditional statements, calculate the value of y for the corresponding value of θ . Once y and θ are known, plot y vs θ .
- 5(b). (i) What text is printed to the screen if the following code snippet is executed? 3×2=6
- ```
for n=1:3
 for m= 1:2:5
 printf("%d %d", n, m);
 if (m==2)
 break;
 endif
 endfor
endfor
```
- (ii) What is wrong with the following code (find at least three mistakes) ?
- ```
for n =1:10
```


(iii) What is wrong with the following code (find at least three mistakes)?

```
S = input ("Enter a text string" );
if (S= "Hi")
disp ( " You entered Hi" );
elseif
disp ( "You did not enter Hi" );
end
```

5(c). What will be printed to the screen if the following commands are executed? 06

a) printf ("Hello World \r"); (b) printf (" Hello World \b\n");
c) printf ("Hello World \b\t\n"); (d) printf ("% d\n", 2);
e) printf ("% f\n", 2); (f) printf ("% e\n", 2);

5(d). Which of the following function definitions are erroneous? Show the errors. 06

(a) $a = fun(x)$ (b) $function\ a = fun(x\ y\ z)$
 $a = sin(x)$ $a = x + y + z;$
endfunction endfunction

(c) $function\ fun()$ (d) $function\ [a, b] = fun(x)$
printf ("Hello World \n"); $a = sin(x);$
end endfunction

(e) # my sine function (f) $function\ a = fun(x, y)$
 $function\ a = fun(x)$ $b = sqrt(x);$
#Usage : $a = fun(x)$ $c = sqrt(y) * b;$
 $a = sin(x);$ end
endfunction

5(e) What are function handle, function function, subfunction, and whos? 07

6(a). Write an Octave code for creating a 5×5 random matrix. Use built-in function to find eigenvectors and eigenvalues of that matrix. Also check if the eigenvectors are linearly independent and span $\mathbb{R}^5$. 15

6(b). Create three row vectors: 10

$a = [3\ -1\ 5\ 11\ -4\ 2],\ b = [7\ -9\ 2\ 13\ 1\ -2]$
 $c = [-2\ 4\ -7\ 8\ 0\ 9]$

i) Use these three vectors in an Octave command to create a matrix in which the rows are the vectors a, b, c, and c.

ii) Use these three vectors in an Octave command to create a matrix in which the columns are the vectors b, c, and a.

6(c). What is the geometrical interpretation of eigenvector and eigenvalues of a square matrix. Write down ten axioms of a vector space. 10

7(a). Write code for solving following problems- 15

(i) make surface plot of the function $z = \frac{xy^2}{x^2+y^2}$ for $-1 \leq x \leq 3$ and $1 \leq y \leq 4$

(ii) $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^k}{k!} + \dots$ Use while loop to calculate $e^{1.5}$ using first 20 terms of the series above.

(iii) A epicycloid is given by following:

$x = 13 \cos(t) - 2 \cos(6.5t)$

$y = 13 \sin(t) - 2 \sin(6.5t)$

Plot the cycloid for $0 \leq t \leq 4\pi$

- 7(b). The reciprocal Fibonacci constant φ is defined by the infinite sum: 15
 $\varphi = \sum_{n=1}^{\infty} \frac{1}{F_n}$, where F_n are the Fibonacci number 1, 1, 2, 3, 5, 8, 13, Each element in this sequence of number is the sum of the previous two, Start by setting the first two elements equal to 1, then $F_n = F_{n-1} + F_{n-2}$. Write a MATLAB program in a script file that calculate φ for a given $n = 50$.
- 7(c). Write short notes on: (i) Singular value decomposition. (ii) Changing rank of a matrix. 05
- 8(a). The first derivative $\frac{df(x)}{dx}$ of the a function $f(x)$ at a point $x = x_0$ can be approximated with two point central difference formula: 20

$$\frac{df(x)}{dx} = \frac{f(x_0 + h) - f(x_0 - h)}{2h}$$
Write an Octave function $dfdx = \text{centraldiff}(\text{Fun}, x_0)$ where Fun is the name of the function that is passed into centraldiff, and x0 is the point where the derivative is calculated. Use $h = \frac{x_0}{100}$ in the two-point central difference formula. Use centraldiff to calculate-
(i) $f(x) = x^3 e^{2x}$ at $x_0 = 0.6$
(ii) $f(x) = \frac{3x}{x^2}$ at $x_0 = 2.5$
- 8(b). Write a user –defined function that adds or subtracts two polynomials of any order. Name the function P= Polyadd (P1, P2, operation). The first two input arguments P1 and P2 are the vectors of the coefficients of the two polynomials. (If the two polynomials are not of the same order the function adds the necessary zero elements to the shorter vector). The third input argument ‘operation’ is a string that can be either ‘add’ or ‘sub’, for adding or subtracting the polynomials respectively. The output argument is the resulting polynomial. Use the function to add and subtract the following
 $f_1(x) = 2x^6 - 3x^4 - 9x^3 + 11x^2 - 8x + 4$ and $f_2(x) = 5x^3 + 7x - 10$ 15